

REVISION HISTORY

Version No.	Date	Originator	Summary of Changes
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Project Overview

The objective of the Mobile Tool Assistant project was to deliver a proof-of-concept mobile robotic prototype capable of autonomously following a designated worker and transporting a tool payload of up to 25 kg in an indoor industrial environment.

During initial supervisor meetings (October 2025), the team made strategic decisions to eliminate features such as an auto-docking/return-to-base function to ensure the project remained feasible within the timeframe.

- Budget Constraint: \$1000 CAD maximum.
- Schedule Constraint: Completion within the academic year (November 2025 – March 2026).
- Scope Boundaries: Operations restricted to flat, single-level indoor environments at safe walking speeds (≤ 5 km/h). Vision-based and BLE-based tracking alternatives were scoped out in favor of UWB (Ultra-Wideband) tracking to balance cost and accuracy.

Team Governance and Roles

A four-member engineering team executed the project. Project governance relied on a structure in which tasks were divided by subsystem requirements, and integration was handled collaboratively.

Table 1: Individual Member Contribution Summary

Member	Total Hours	Key Contributions
Aneek Mohamed Ruksan	180	UWB User Tracking, UI/User Node
Aditya Ramakrishnan	120	Motor Control Algorithm, Power Management System, Obstacle Avoidance, Chassis Design and Printing, System Refining and Improvement, System Assembly
Ethan D’Almeida	70	UWB calibration support and testing, documentation, obstacle detection, and avoidance improvements
Chidubem Emeka-Nwuba	75	Obstacle Detection Subsystem Assembly and Integration, Software architecture: implementation, management, and subsystem task integration

Project Monitoring and Supervisor Meetings

To ensure the project stayed on track, the team held regular design and status meetings with the Project Supervisor, Chandler Janzen. These meetings served as the primary way we tracked project progress and issue tracking.

Summary of Key Meeting Outcomes:

- **Oct 03, 2025 (Requirements Phase):** Focused on defining testable, quantifiable requirements. Identified the risk of scope creep and decided to strictly limit the operational environment to indoors with a 5 km/h speed limit.
- **Oct 21, 2025 (System Alternatives Phase):** Addressed potential safety hazards (stairs/holes). Decided to minimize mechanical design details and remove the auto-docking requirement to reduce overall scope and focus on electronic/control feasibility.
- **Nov 04, 2025 (Project Planning Phase):** Clarified testing metrics (e.g., using a decibel meter for noise constraints).
- **Nov 25, 2025 (Detailed Design Phase):** Established documentation standards, requiring every requirement to map directly to a design element with clearly defined subsystem inputs/outputs.
- **Feb 26, 2026 (Execution & Testing Phase):** Conducted a prototype demonstration. Logged critical issues regarding UWB module calibration inconsistencies, motor alignment, and chassis mounting.

Financial and Resource Tracking

The project was controlled to remain under the \$1000 CAD constraint. Financial planning involved estimating material costs early in the design phase and adjusting based on part availability.

Table 2: Bill of Materials

Part Name	Quantity	Cost (in CAD)
STM32 Dev Board	1	\$42.65
DWM1000 Module (UWB)	3	\$122.91
VL531X Module (ToF)	3	\$64.77
L298N Motor Driver	1	\$11.78
ESP8266	1	\$4.05
TT DC Gear Motor	2	\$4.00
Buck Converter Module	1	\$21.99
Rechargeable Li-ion Battery	2	\$21.40
Small Yellow Button	1	\$0.27
Total		\$293.82

Development Resources

- **Hardware:** Multimeter, Soldering Iron, BambuLab P1-S (for chassis).
- **Software:** STM32Cube IDE v1.19.0, Arduino IDE v2.3.7, Serial Debug Assistant, GitHub (version control).

Scheduling, Monitoring, and Control

The project timeline experienced significant deviations from the initial baseline plan. By actively monitoring the schedule, the team was able to apply mitigation strategies such as simplifying the UI and running tasks in parallel to ensure final delivery.

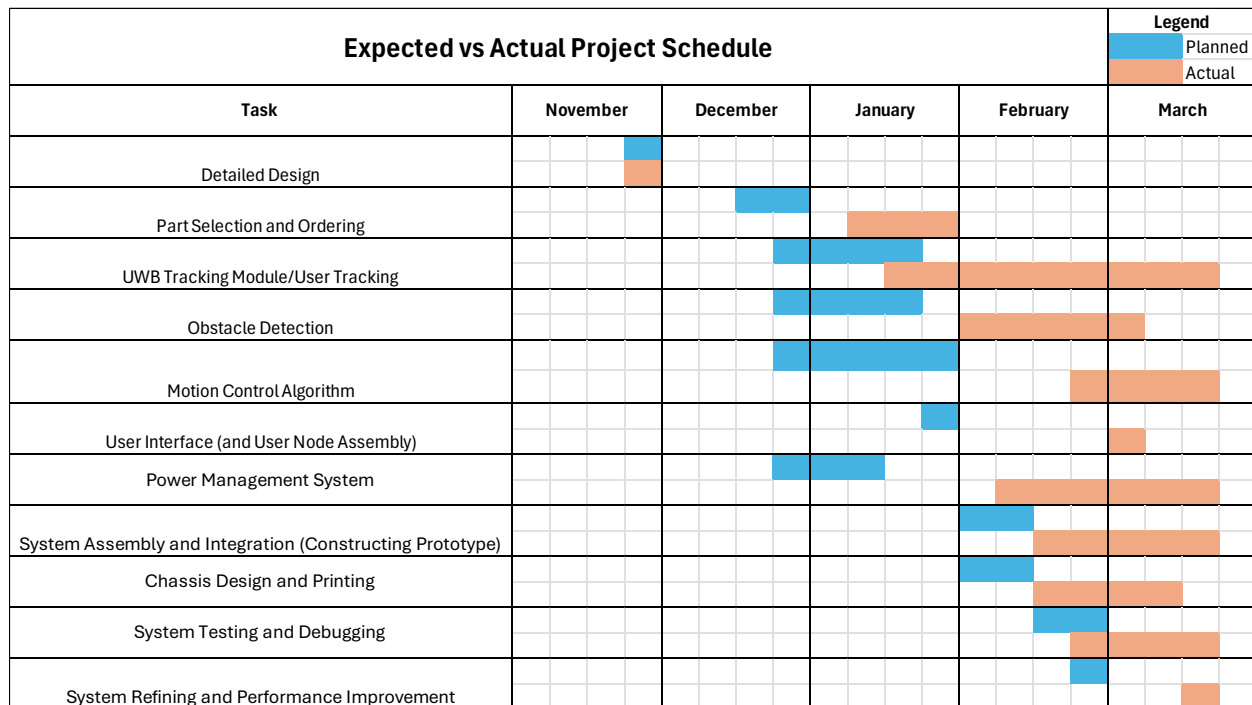


Figure 1: Gantt Chart of Expected v/s Actual Project Schedule

Table 3: Description of Planned v/s Actual Hours and Deadlines

Task ID	Task Name	Planned Hours	Planned Deadline	Actual Hours	Actual Completion	Comments
1	Detailed Design	20h	27 th Nov, 2025	20h	27 th Nov, 2025	On time
2	Part Selection and Ordering	10h	1 st Dec, 2025	25h	27 th Jan, 2026	Delayed due to redesigns and part availability
3	UWB Tracking Module	40h	21 st Jan, 2026	175h	12 th Mar, 2026	Required extra time for antenna delay calibration, DS-TWR algorithm.
4	Obstacle Detection	45h	21 st Jan, 2026	45h	4 th Mar, 2026	Integrating multiple ToF sensors
5	Motion Control Algorithm	21h	30 th Jan, 2026	43h	24 th Mar, 2026	Delayed due to motor alignment and integration with UWB
6	User Interface	20h	30 th Jan, 2026	5h	13 th Mar, 2026	Saved time here, but overall, UWB cascading delay
7	Power Management System	24h	21 st Jan, 2026	24h	20 th Mar, 2026	UWB cascading delay
8	System Assembly and Integration	30h	15 th Feb, 2026	30h	23 rd Mar, 2026	UWB cascading delay
9	Chassis Design and Printing	15h	15 th Feb, 2026	15h	19 th Mar, 2026	UWB cascading delay
10	System testing and debugging	37h	28 th Feb, 2026	40h	24 th Mar, 2026	Delay to ensure stability
11	System Refining and Improvement	16h	28 th Feb, 2026	37h	24 th Mar, 2026	More time to refine the motor movements and obstacle detection algorithm

Risk Issue and Management

Table 4: Identified Risks and corresponding Mitigation Strategies

Identified Risk	Impact	Occurrence/Mitigation Strategy
UWB (new technology)	High	UWB implementation took more hours than planned since it was the new technology we were working with, which caused a chain of long delays. Mitigation: Ethan supported Aneek with calibration while other team members worked on power and obstacle detection in parallel to continue project progress.
Component Availability and Delivery	Medium	Parts Selection took longer than expected due to availability. Mitigation: Critical components were identified and ordered early.
System Integration Issues	High	Integration of motion control and user tracking subsystems took longer than expected due to motor alignment and algorithm issues. Mitigation: Extensive testing was done for each subsystem to fix any issues before fully integrating both systems.
Schedule Overruns	High	Occurred

Project Closure and Lessons Learned

The project successfully delivered a working proof-of-concept that met the core functional requirements, passing final testing for follow distance ($1\text{m} \pm 0.5\text{m}$ radius), speed (up to 5 km/h), payload capacity (25kg), and obstacle detection (within 150cm).

Throughout the development of the Mobile Tool Assistant, the team identified several key lessons and areas for architectural improvement. Most notably, the team drastically underestimated the learning curve and complexity of implementing custom Ultra-Wideband (UWB) tracking from scratch, highlighting the need in future projects to allocate significantly more schedule buffer for researching and calibrating unfamiliar communication protocols. On the hardware front, relying on jumper wires and protoboards helped with rapid prototyping but introduced signal instability and loose connections during physical testing; future iterations must transition to custom PCBs to ensure electrical reliability and mechanical robustness. Additionally, the prototype's sensor, utilizing front-facing Time-of-Flight (ToF) sensors and two UWB anchors, limited the robot to a 180° field of view, leaving it vulnerable if the user moved behind it. A production model would require a 360° LIDAR sensor and a third rear UWB anchor to achieve full coverage. Finally, upgrading the power system from standard rechargeable AA batteries to an integrated, industry-standard lithium-ion battery pack with a dedicated Battery Management System will be essential to improve energy density and long-term operational performance.